🔹 What is a Schema (Simple Explanation)

A **schema** is the **structure or blueprint of a database**.

👉 It tells:

- What tables exist
- What columns each table has
- Data types of columns
- Relationships between tables

🧠 Think Like This

👉 If **database = folder**
👉 Then **schema = design of that folder**

Or even better:

- Database = House
- Schema = House blueprint (plan)
- Tables = Rooms inside the house

🔹 Example

When you write:

```
CREATE TABLE employees (
  id INT,
  name VARCHAR(50),
  salary DECIMAL(10,2)
);
```

👉 This structure itself is the **schema** of the employees table.

🔹 Schema vs Data

Schema	Data
Structure (design)	Actual values
Columns, types	Rows (records)
Fixed	Changes often

👉 Example:

- Schema → id INT, name VARCHAR(50)
- Data → (1, 'Indra')

🔹 **Important Point (MySQL Specific)**

In **MySQL, schema and database are almost the same thing**.

👉 These two commands mean the same:

CREATE DATABASE company;

CREATE SCHEMA company;

🔥 **Why Schema Matters**

Without schema:

- Data becomes messy
- No consistency
- Hard to query

With schema:

- Clean structure
- Faster queries
- Easy to manage large data

🚀 **Simple One-Line Definition**

👉 **Schema = how your data is organized before you store it**

📌 **What is SHOW in MySQL?**

In **MySQL**,
👉 SHOW is a command used to **display metadata** (info about databases, tables, etc.)

🔹 **1. Show all databases**

SHOW DATABASES;

👉 Lists all databases in your system

🔹 **2. Select a database**

USE testdb;

🔹 **3. Show all tables**

SHOW TABLES;

👉 Shows tables inside the selected database

🔹 **4. Show table structure**

DESCRIBE employees;

or

SHOW COLUMNS FROM employees;

👉 Shows:

- Column names
- Data types
- Keys

🔹 **5. Show table creation query**

SHOW CREATE TABLE employees;

👉 Very useful in real jobs (debugging, migration)

🔹 **6. Show current database**

SELECT DATABASE();

🔹 **7. Show MySQL version**

SELECT VERSION();

🔹 **Real Example Flow (Do this step-by-step)**

SHOW DATABASES;

```
USE testdb;

SHOW TABLES;

DESCRIBE employees;
```

👉 If you run this, you'll understand your database structure clearly.

🧠 **Why SHOW is Important**

In real work:

- You join a project → don't know structure
- You use SHOW to explore database

👉 This is how professionals navigate unknown systems

⚠️ **Reality Check**

Beginners ignore this, but:
👉 Good engineers always **inspect before querying**

🎯 **Interview Tip**

👉 "SHOW commands are used to retrieve metadata such as databases, tables, and schema details in MySQL."

Good—this is a simple but **very important command**. If you don't use it correctly, your queries won't work.

📌 **What is USE in MySQL?**

In **MySQL**,

👉 USE is a command to **select a database** so you can work inside it.

🔹 **Simple Definition**

👉 "USE tells MySQL which database you want to work with."

🔹 **Syntax**

USE database_name;

🔹 **Example**

USE testdb;

👉 Now MySQL understands:
“All queries should run inside testdb”

🔹 **Why USE is Needed**

Imagine you have multiple databases:

- testdb
- company
- school

👉 MySQL gets confused unless you specify one

So you must do:

USE company;

🔹 **Real Flow (Very Important)**

SHOW DATABASES;

USE testdb;

SHOW TABLES;

SELECT * FROM employees;

👉 This is how you actually work in real life

🧠 **Simple Analogy**

- Database = folder 📁
- USE = opening that folder

If you don’t open the folder, you can’t access files.

⚠️ **Common Mistake**

❌ Running query without selecting database:

```
SELECT * FROM employees;
```

👉 Error: “No database selected”

✔️ Fix:

```
USE testdb;
```

🎯 **Interview Answer**

👉 “The USE command is used to select a specific database so that all subsequent queries are executed within that database.”

🔹 **1. Create a Database**

A **database** is like a container that holds your data.

```
CREATE DATABASE company_db;
```

👉 This creates a database named company_db.

🔹 **2. Use the Database**

Before creating tables, you must select the database:

```
USE company_db;
```

🔹 **3. Create a Table**

A **table** is where actual data is stored (like an Excel sheet but structured).

```
CREATE TABLE employees (
  id INT PRIMARY KEY,
  name VARCHAR(50),
  age INT,
  salary DECIMAL(10,2)
);
```

🔍 **Breakdown:**

- id INT PRIMARY KEY → unique identifier
- name VARCHAR(50) → text (max 50 characters)
- age INT → number
- salary DECIMAL(10,2) → decimal values like 50000.50

🔹 **4. Check Your Table**

```
SHOW TABLES;

DESCRIBE employees;
```

🔹 **5. Insert Data**

```
INSERT INTO employees (id, name, age, salary)

VALUES (1, 'Indra', 22, 50000);
```

🔹 **6. View Data**

```
SELECT * FROM employees;
```

💡 **Real Understanding (Important)**

Think like this:

- Database → Folder
- Table → Excel sheet
- Rows → Records
- Columns → Fields

```
SELECT DATABASE();
```

🔹 **What does SELECT DATABASE() do?**

👉 It **returns the name of the currently selected database**.

🧠 **Simple Meaning**

When you run:

USE company_db;

You are telling MySQL:
👉 “Work inside this database”

Then if you run:

SELECT DATABASE();

👉 MySQL replies:

company_db

🔹 **Why is this useful?**

Sometimes you:

- Forget which database you're using
- Work with multiple databases
- Switch frequently

👉 This command helps you **confirm your current database**

🔍 **Example**

CREATE DATABASE test_db;

USE test_db;

SELECT DATABASE();

👉 Output:

test_db

⚠️ **Important Case**

If you didn’t select any database:

SELECT DATABASE();

👉 Output:

NULL

Meaning:
👉 “No database is currently selected”

🚀 **One-Line Understanding**

👉 SELECT DATABASE() = “Which database am I using right now?”

🔹 **Delete a Database in MySQL**

👉 The command is:

DROP DATABASE database_name;

🔹 **Example**

DROP DATABASE company_db;

👉 This will:

- Delete the database
- Delete **all tables inside it**
- Delete **all data permanently**

⚠️ **Very Important (Don’t Ignore)**

❌ Once deleted → **you cannot recover it**

This is not like deleting a file from recycle bin — it’s **gone forever** unless you have a backup.

🔹 **Safe Way (Recommended)**

Before deleting, check:

SHOW DATABASES;

Then delete:

DROP DATABASE company_db;

🔹 **Avoid Errors (Better Practice)**

DROP DATABASE IF EXISTS company_db;

👉 This prevents error if the database doesn't exist.

🧠 **Simple Understanding**

👉 DROP DATABASE = "Delete the entire folder and everything inside it"

🚀 **Pro Tip (Think Like a Data Engineer)**

Never casually run DROP DATABASE in real projects.

Instead:

- Take backup first
- Double-check environment (dev vs production)
- Use with caution

A table is a collection of related data held in a table format within a database .

🔹 **1. Create a Table in MySQL**

👉 Basic syntax:

```
CREATE TABLE table_name (
   column1 datatype,
   column2 datatype,
   column3 datatype
);
```

🔹 **Example**

```
CREATE TABLE students (
   id INT PRIMARY KEY,
   name VARCHAR(50),
   age INT,
```

```
  marks INT
);
```

🔍 **What this means:**

- students → table name
- id INT PRIMARY KEY → unique ID (no duplicates)
- name VARCHAR(50) → text up to 50 characters
- age INT → integer values
- marks INT → student marks

👉 This is your **table structure (schema)**

🔹 **2. What is DESCRIBE?**

👉 The DESCRIBE command shows the **structure of a table**

🔹 **Syntax**

```
DESCRIBE table_name;
```

🔹 **Example**

```
DESCRIBE students;
```

🔹 **Output (What You'll See)**

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	
name	varchar(50)	YES		NULL	
age	int	YES		NULL	
marks	int	YES		NULL	

🔍 **Meaning of Each Column**

- **Field** → column name
- **Type** → datatype (INT, VARCHAR, etc.)

- **Null** → can it store NULL values?
- **Key** → primary key (PRI)
- **Default** → default value
- **Extra** → extra info (like auto_increment)

🧠 **Simple Understanding**

👉 CREATE TABLE = build the structure
👉 DESCRIBE = inspect the structure

🚀 **Real-Life Analogy**

- CREATE TABLE → Designing an Excel sheet
- DESCRIBE → Looking at column headers and formats